

Session 8 Quiz - Large Language Models (LLMs)

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57004]

Fill in the blank: A Language Model is a type of AI that is designed to predict the next ____.

- A. feeling
- B. word
- C. picture
- D. memory

Correct Answer: B. word

Explanation: The core function of a language model is to determine which word most likely follows a given sequence based on learned patterns. This is similar to how your phone suggests the next word when you type a text message.

Q2 [Type: MCQ | Difficulty: Easy | QID:57005]

Fill in the blank: AI doesn't actually predict whole words; it predicts pieces of language called ____.

- A. syllables
- B. paragraphs
- C. bytes
- D. tokens

Correct Answer: D. tokens

Explanation: Tokens are the LEGO blocks of AI language, representing small pieces or segments of words. By breaking language into tokens, AI can handle new or unusual words by combining known pieces.

Q3 [Type: MCQ | Difficulty: Easy | QID:57009]

Fill in the blank: Large Language Models are trained by reading ____ of sentences from the internet.

- A. millions
- B. dozens
- C. billions
- D. hundreds

Correct Answer: C. billions

Explanation: The Large in LLM refers to the massive scale of data used, which includes billions of webpages, books, and conversations. This huge amount of text allows the model to learn rich language patterns.

Q4 [Type: MCQ | Difficulty: Easy-Medium | QID:57010]

Fill in the blank: An AI can sound very sure of itself even when its answer is totally ____.

- A. correct
- B. quiet
- C. human
- D. wrong

Correct Answer: D. wrong

Explanation: LLMs prioritize pattern matching over truth, which means they can state incorrect facts with total confidence. This is the confidently wrong weakness mentioned in the text.

Q5 [Type: MCQ | Difficulty: Easy | QID:57012]

Fill in the blank: The text compares LLMs to advanced ____ because they repeat patterns without understanding meaning.

- A. parrots
- B. monkeys
- C. computers
- D. students

Correct Answer: A. parrots

Explanation: Parrots repeat sounds that work in certain situations without understanding the concepts behind the words. Similarly, LLMs repeat language patterns from their training data without true understanding.

Q6 [Type: True/False | Difficulty: Easy | QID:57013]

True or False: Large Language Models (LLMs) have feelings and can remember your past conversations forever.

- A. True
- B. False

Correct Answer: B. False

Explanation: The source material explicitly states that LLMs have no feelings, no memories, and can get confused by long conversations. They do not remember past chats unless the conversation is within their limited context window.

Q7 [Type: True/False | Difficulty: Easy | QID:57014]

True or False: ChatGPT and Gemini often give different answers because they were trained on different data and have different settings.

A. False

B. True

Correct Answer: B. True

Explanation: Different companies use different datasets and personality settings, leading to unique strengths and weaknesses for each AI. That is why you were asked to compare both tools side by side.

Q8 [Type: True/False | Difficulty: Easy-Medium | QID:57015]

True or False: When an AI finishes a sentence, it is using a master plan to determine the entire paragraph at once.

A. False

B. True

Correct Answer: A. False

Explanation: AI builds sentences one token at a time, picking the most likely next piece based on what came right before it. This is like dropping breadcrumbs, not following a master plan for the whole paragraph.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:57016]

True or False: Because LLMs learn from human writing, they can sometimes repeat human biases or unfair ways of thinking.

- A. True
- B. False

Correct Answer: A. True

Explanation: LLMs reflect the data they are given, which includes any biases present in the billions of sentences they have read. This is mentioned as one of the weaknesses of LLMs.

Q10 [Type: True/False | Difficulty: Easy | QID:57018]

True or False: A language model understands the meaning of the words it uses just like a human does.

- A. True
- B. False

Correct Answer: B. False

Explanation: The text states that your brain understands meaning, while the model just matches patterns it has seen before. LLMs do not have true understanding or consciousness.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:57019]

What is the primary reason an LLM sounds so human when you talk to it?

- A. There is a person on the other end typing the responses.
- B. It is an imitation based on billions of words written by people.
- C. It uses a special camera to see your face and mimic you.
- D. It has developed its own personality and soul.

Correct Answer: B. It is an imitation based on billions of words written by people.

Explanation: Because the training data is 100% human created, the AI's output naturally mirrors human style and emotion. It is a really, really good imitation, not actual humanity.

Q12 [Type: MCQ | Difficulty: Medium | QID:57020]

When training an AI, what happens after the model makes a guess about a missing word?

- A. It compares its answer to the original and updates its internal settings.
- B. It asks the user for the correct answer immediately.
- C. It deletes the sentence and starts over from scratch.
- D. It gives up and tries to generate a picture instead.

Correct Answer: A. It compares its answer to the original and updates its internal settings.

Explanation: By checking if its guess was right or wrong, the AI updates its weights to become more accurate next time. This is the check and learn step in the training process.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:57021]

Which of these tasks is an LLM most likely to struggle with?

- A. Translating a sentence into Spanish.
- B. Summarizing a long article about space.
- C. Providing 100% accurate info on a very recent event.
- D. Writing a poem about a robot.

Correct Answer: C. Providing 100% accurate info on a very recent event.

Explanation: LLMs often have outdated info or mix old data with new, making them unreliable for current events. They can translate, summarize, and write poetry reasonably well.

Q14 [Type: MCQ | Difficulty: Medium | QID:57022]

How do 'tokens' help an AI handle a word it has never seen before?

- A. It searches the entire live internet for a definition.
- B. It can break the new word into smaller LEGO-like pieces it already knows.
- C. It shuts down to prevent an error.
- D. The AI skips the word and moves to the next one.

Correct Answer: B. It can break the new word into smaller LEGO-like pieces it already knows.

Explanation: By combining known tokens, the AI can construct or understand brand new words it did not specifically learn. For example, moon + light = moonlight.

Q15 [Type: MCQ | Difficulty: Easy-Medium | QID:57023]

What is the best way to handle information given by an AI for a school project?

- A. Double-check the important facts using another source.
- B. Ask the AI if it is lying to see if it admits it.
- C. Copy and paste it because AI is always accurate.
- D. Assume it's wrong because AI cannot answer questions.

Correct Answer: A. Double-check the important facts using another source.

Explanation: Verifying information is essential because LLMs match patterns rather than truly understanding the truth. Always double-check important facts.

Q16 [Type: MCQ | Difficulty: Easy-Medium | QID:57024]

If an AI is like a 'Super Advanced Parrot,' what is it primarily repeating?

- A. Facts that it has memorized from an encyclopedia.
- B. The sounds it hears through the computer microphone.
- C. Whatever the user just typed into the chat box.
- D. Patterns of language that it saw in its training data.

Correct Answer: D. Patterns of language that it saw in its training data.

Explanation: The AI learns which words follow others, mimicking the structure of human language perfectly. It is repeating learned patterns, not memorized facts or live sounds.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:57025]

What did the 'Cat sat on the...' exercise demonstrate about your brain?

- A. Your brain is much slower than a computer.
- B. You only think in percentages all day long.
- C. You are unable to finish a sentence without help.
- D. You can predict words based on patterns you've heard before.

Correct Answer: D. You can predict words based on patterns you've heard before.

Explanation: Like an LLM, your brain uses past experiences to guess what likely comes next in a sentence (e.g., cat sat on the mat or chair). This shows pattern prediction is natural for humans too.

Q18 [Type: MCQ | Difficulty: Medium | QID:57026]

Why might an LLM 'lose track' of a conversation if it goes on for too long?

- A. It gets bored and wants to talk about something else.
- B. It runs out of battery power.
- C. It only likes to answer three questions at a time.
- D. It has a 'context limit' and starts to forget the earliest parts.

Correct Answer: D. It has a 'context limit' and starts to forget the earliest parts.

Explanation: Long conversations exceed the amount of previous text the AI can look at to predict the next word. This context limit causes it to lose track of earlier parts of the conversation.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:57027]

Which of these is NOT something an LLM can actually do?

- A. Help you write a joke about a pizza.
- B. Genuinely care about how your day was.
- C. Brainstorm 10 names for a pet rabbit.
- D. Explain a scientific concept to a child.

Correct Answer: B. Genuinely care about how your day was.

Explanation: Although it can sound caring, the AI has no feelings and does not actually care about anything. It can write jokes, brainstorm names, and explain concepts, but it cannot genuinely care.

Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:57028]

What is the main goal of the 'LLM Detective' homework activity?

- A. To write a full book using only AI.
- B. To find out which AI is the fastest.
- C. To try and make the computer crash with a hard question.
- D. To see how different LLMs handle the same topics and check their accuracy.

Correct Answer: D. To see how different LLMs handle the same topics and check their accuracy.

Explanation: By asking the same questions to different models, you can spot the unique personalities and mistakes of each. The goal is to compare and verify, not to write books, test speed, or crash computers.